INCH-POUND

MIL-DTL-21097/4F
18 August 2005
SUPERSEDING
MIL-C-21097/4E
29 March 1976

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, PRINTED WIRING BOARD COMPOSITE, RECEPTACLE, CONTACT SPACING (.200) ALTERNATE DUAL ROW, TYPE CR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-21097.

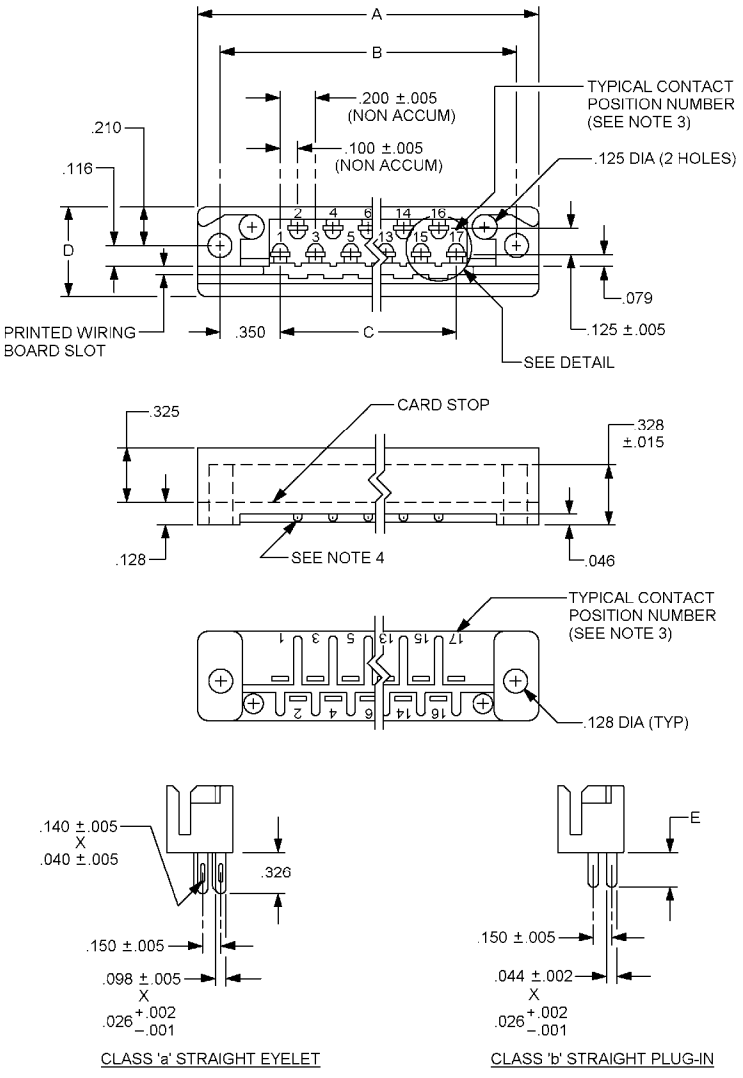
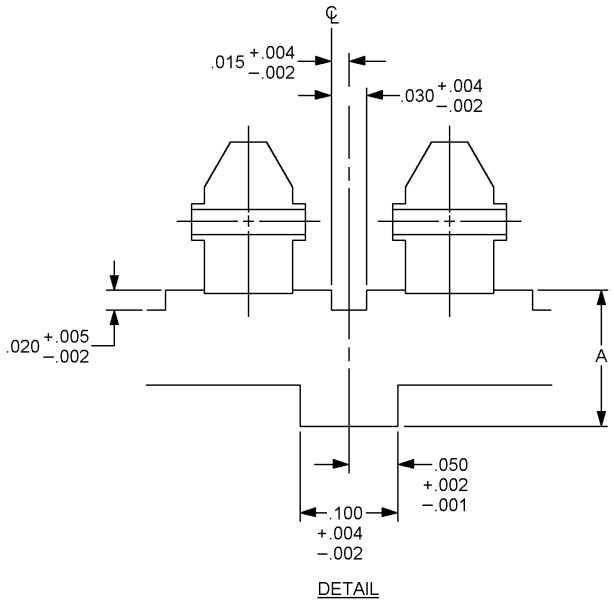


FIGURE 1. Dual row contacts, type CR.

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DETAIL

POLARIZING CAVITY FOR EVEN NUMBER OF CONTACT POSITIONS

Inches	mm	Inches	mm
.001	.03	.098	2.49
.002	.05	.100	2.54
.004	.10	.116	2.95
.005	.13	.125	3.18
.015	.38	.128	3.25
.020	.51	.140	3.56
.026	.66	.150	3.81
.030	.76	.200	5.08
.040	1.02	.210	5.33
.044	1.12	.325	8.26
.046	1.17	.326	8.28
.050	1.27	.328	8.33
.079	2.01	.350	8.89

Board guide	A dim + .002 (.05) - .004 (.10)
.06 (1.52) .09 (2.29)	.119 (3.02) .150 (3.81)

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ± .010 (.25 mm) for three place decimals.
3. Contacts and terminals shall be designated on the connector using numerals in the locations shown.
4. The shape of the barriers is optional except that barriers shall not protrude beyond the plane of the underside of the mounting flange.
5. Slot shall accept the following thickness printed wiring boards:
   Size 2 - .054 (1.37 mm) to .071 (1.80 mm)
   Size 3 - .086 (2.18 mm) to .103 (2.62 mm)
   Slot shall prevent entry of printed wiring board having the following thickness:
   Size 2 - .084 (2.13 mm)
   Size 3 - .113 (2.87 mm)
6. Metric equivalents are given for information only .
7. Millimeters are in parentheses.

FIGURE 1. Dual row contacts, type CR – Continued.

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TABLE I. Connector, electrical, composite, receptacle, type CR, mounting provision A.

Dash number	Style	Class	Size	Dimensions (inches) 1/ 2/				
				A (Maximum)	B ±0.015 (0.38)	C ±0.005 (0.13)	D ±0.010 (0.25)	E ±0.010 (0.25)
01		a	2					
02		a	3					
03		b	2					.123 (3.12)
04	17	b	2	2.570 (65.28)	2.300 (58.42)	1.600 (40.64)	0.531 (13.49)	.154 (3.91)
05		b	2					.186 (4.72)
06		b	3					.123 (3.12)
07		b	3					.154 (3.91)
08		b	3					.186 (4.72)
09		a	2					
10		a	3					
11		b	2					.123 (3.12)
12	23	b	2	3.170 (80.52)	2.900 (73.66)	2.200 (55.88)	0.531 (13.49)	.154 (3.91)
13		b	2					.186 (4.72)
14		b	3					.123 (3.12)
15		b	3					.154 (3.91)
16		b	3					.186 (4.72)

See footnotes at end of table.

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TABLE I. Connector, electrical, composite, receptacle, type CR, mounting provision A – Continued

Dash number	Style	Class	Size	Dimensions (inches) 1/ 2/ A (Maximum)	B ±0.015 (0.38)	C ±0.005 (0.13)	D ±0.010 (0.25)	E ±0.010 (0.25)
17		a	2					
18		a	3					
19		b	2					.123 (3.12)
20	29	b	2	3.770 (95.76)	3.500 (88.90)	2.800 (71.12)	0.531 (13.49)	.154 (3.91)
21		b	2					.186 (4.72)
22		b	3					.123 (3.12)
23		b	3					.154 (3.91)
24		b	3					.186 (4.72)
25		a	2					
26		a	3					
27		b	2					.123 (3.12)
28	35	b	2	4.370 (111.00)	4.100 (104.14)	3.400 (86.36)	0.531 (13.49)	.154 (3.91)
29		b	2					.186 (4.72)
30		b	3					.123 (3.12)
31		b	3					.154 (3.91)
32		b	3					.186 (4.72)
33		a	2					
34		a	3					
35		b	2					.123 (3.12)
36	41	b	2	4.970 (126.24)	4.700 (119.38)	4.000 (101.6)	0.531 (13.49)	.154 (3.91)
37		b	2					.186 (4.72)
38		b	3					.123 (3.12)
39		b	3					.154 (3.91)
40		b	3					.186 (4.72)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

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REQUIREMENTS

Dimensions and configuration: See figure 1 and table I.

Polarizing keys: Part or Identifying Number (PIN). M21097/11-03 or M21097/11-04. A key is inserted into the card slot at even contact numbers, 2, 4, 6, etc.

Mating male adapters: Shall be in accordance with MIL-DTL-21097/5.

Finish: Gold plating in accordance with ASTM B488, type II, code C, class 1.27, over nickel plating in accordance with SAE-AMS-QQ-N-290, class 2, 30 to 150 microinches.

PIN: M21097/4-(dash number from table I).

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-21097, this document references the following:

MIL-DTL-21097/5
ASTM B 488
SAE-AMS-QQ-N-290

CONCLUDING MATERIALS

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-4505-012)

Review activities:
Army – AR, AT, AV, CR4, MI
Navy – AS, MC, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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